Do Environmental Provisions in Trade Agreements Reshape Export Composition?

Evidence from the Universe of Chinese Customs Transactions, 2000–2015

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Abstract

Environmental Provisions (EPs) have been incresignly embedded in Preferential Trade Agreements (PTAs) over the last two decades, with the expectation that they will play an even more important role in future agreements. However, empirical evidence on their trade effects relies almost entirely on aggregate bilateral flows. This paper asks whether EPs can shift export composition towards green goods and away from dirty goods, using a firm-product-destination-year panel for the universe of Chinese customs transactions spanning from 2000 to 2015. A triple-difference design allows to compare green and dirty products againts neutral counterparts within each firm-destination-year combination, with firm-destination-year fixed effects accounting for everything constant within each firm-market-year triple. The results indicate that there is no effect on the green margin, with the dirty margin being less clear. While the inference on the green margin result is robust to wild cluster bootstrap and permutation inference, that for the dirty margin is not. What emerges is that the content of environmental provisions matter more than their mere presence. PTAs signed by China across $2000 - 2015$ were mostly cooperation-language only, which makes our results consistent with existing literature, where significant results are driven by specific and enforceable clauses, rather than loose language.

1 Introduction

Over the last decades, the number of preferential trade agreements (PTAs) has increased substantially, and many now include environmental provisions (EPs) or other forms of environmental language (Mattoo, Rocha, and Ruta, 2020; Morin, Pauwelyn, and Hollway, 2018). Because climate change is likely to become an increasingly important political concern, and given the evidence that global trade may play a role in environmental decay (Zhang et al., 2017) environmental content in trade agreements is expected to feature more frequently and more prominently in trade policy, as may serve as a possible tool to address challenges posed by climate change (Frankel 2009). Accordingly, the relationship between trade and the environment has attracted growing attention in the literature (Copeland, Shapiro, and Taylor, 2022), yet whether environmental provisions in trade agreements have a genuine effect on trade flows is still an open empirical question (Felbermayr, Peterson, and Wanner, 2025; Gutsch et al., 2024). This paper contributes to this literature by asking whether the environmental provisions included in preferential trade agreements signed by China between 2000 and 2015 shifted the composition of its exports toward greener products and away from dirtier ones, or whether those provisions left trade flows largely unaffected.

Most of the existing literature on trade effects of environmental provisions in trade agreements comes from studies relying on aggregate bilateral trade flows, which makes it difficoult to understand whether EPs cause a compositional shift in trade goods. In what the authors claim to be the first article to investigate trade effects of environmental provisions in PTAs, Brandi et al. (2020) find that EPs reduce dirty export and increase green exports when regulations are specific and enforcable, especially for developing countries. On the other hand, Berger et al. (2020) use a gravity panel to show that membership in PTAs with more environmental provisions is associated with lower bilateral exports overall, with the effect being concentrated in South-North trade flows and not robust to PPML. Specific and enforcable clauses are those that drive the effect also in Abman, Lundberg, and Ruta, 2024, who find that forest preservation and biodiversity clauses in PTAs reduce deforestation in partner countries relative to those that do not include such clauses. Taken together, the evidence suggests that it is the content of the specific provisions and chapters that matters, rather than they the mere presence in trade agreements. This paper brings the question to the largest exporter in the world, asking whether the environmental provisions in China's PTAs between 2000 and 2015 had any effect on the composition of its exports, and whether the content of those provisions matters for trade flows.

Trying to answer this empirical question comes with some challenges. The first and most obvious is that countries do not sign trade agreements at random, and the signing of trade agreements is likely to be correlated with pre-existing relationship of some sort. Therefore, any comparison of export flows to PTA partners versus non-partners risks to confound the effect of environmental provisions with pre-existing differences in trade relationships. A second issue is that aggregate bilateral trade flows cannot distinguish a reallocation of export composition toward greener goods from an expansion or contration of trade that happens to coincide with agreement entry into force. To address the empirical question, this study uses the universe of Chinese custom transactions between 2000 and 2015, a 45-million observation firm-product-destination-year panel dataset with granularity at the HS6 product level, which allows to track the same firm exporting the same product to multiple destinations simultaneously, and to observe how those flows diverge when one destination is covered by a preferential agreement and another is not. This design allows to isolate the effect of environmental provisions on trade composition from other confounding factors.

Over the study period, China signed 14 PTAs with 25 destination economies, with environmental content ranging from a handful of generic cooperation clauses to dedicated chapters. Data on the content of the environmental provisions come from two independent sources, the World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) and the TREND database (Morin, Pauwelyn, and Hollway, 2018). The former provides a count of environmental-law provisions in force with each destination at each point in time, while the latter provides a more detailed coding of the content of those provisions, including sub-indices covering areas such as green market access and enforcement stringency. Using two independent codings guards against idiosyncrasies of either scheme and allows to check for robustness of the results. We identify green goods on the basis of the OECD Combined List of Environmental Goods (CLEG) (Sauvage, 2014), while dirty goods are identified through a standard pollution-intensity classification that maps polluting industries to HS6 product codes, as developed in Mani and Wheeler, 1998. Everything else serves as the neutral comparison group.

These three product categories, green, dirty and neutral, represents the founding basis of the empirical strategy. The empirical design applied in this study is a comparison of how the export of green and dirty products react to environmental provisions, relative to neutral products, within the same firm, destination and year. A triple-difference with a rich set of fixed effects, including firm-destination-year fixed effects, absorbs everything that is constant within a firm's relationship with a specific destination market in a given year, including the agreement itself and any aggregate demand or supply shocks at the bilateral level. Because EP enter into force with the agreement itself and rarely change their depth or content afterwards, EP depth and PTA depth are highly correlated, as agreements with more EP content tend to be also deeper agreements overall. Therefore, all specifications control for PTA depth separately, so that the coefficient on environmental provisions reflects proper environmental content. In this design, identification comes from the differential sensitivity of product types to environmental provisions green and dirty products against neutral products, which are used as a within-cell control since they are not expected to be affected by environmental provisions as much as green and dirty products are.

The results indicate that there is no effect on the green margin, so the export of green goods relative to neutral ones is not affected by the depth of environmental provisions in China's PTAs. This result is stable across six different estimation designs that vary the unit of observation, the measurement instrument, and the comparison group. The baseline uses the full firm-product-destination-year panel; a collapsed specification aggregates the same data to HS6-destination-year cells; and both are replicated using the TREND provision count as an alternative measure of EP depth to that offered by the World Bank Deep Trade Agreements database. Further robustness checks restrict the sample to CEM-matched destinations and to the common-support overlap region. Across all specifications, we find no statistically significant effect of environmental provisions on the export of green goods relative to neutral ones. Furthermore, these results are robust to wild cluster bootstrap inference (Cameron, Gelbach, and Miller 2008) and permutation inference (Young, 2019). On the other hand, results for the dirty margin are less clear-cut. While we find no effect in the full panel, the collapsed panel shows a negative and significant coefficient, suggesting that while EPs may not change firm-specific behaviour, they may reduce the overall market-level export of dirty goods relative to neutral ones. However, this result does not survive more robust inference. Wild cluster bootstrap raises the p-value from below 0.001 to 0.07; a permutation test, which asks whether the same pattern would emerge by chance if EP depth were randomly reassigned across destinations, raises it further to 0.28.

The findings show that environmental provisions in China's PTAs between 2000 and 2015 had no measurable effect on export composition at the firm-product-destination-year level. This result, however, should not be taken as concluding evidence that EPs in trade agreements have no effect on trade flows at all, but rather reflects the nature of the specific provisions themselves. During the period under analysis, the environmental content of China's agreements consisted almost entirely of soft cooperation language, rather than specific and enforcable clauses enriched with dispute settlement mechanisms. The types of clauses that have been shown to produce significat trade outcomes in other contexts, such as tariff concessions on environmental goods and binding emissions or process standards, are largely absent from these agreements. In this sense, the null result is not a puzzle, but rather the expected outcome given the existing literature's own heterogeneity results, applied to a setting where the provisions lack the specificity and enforceability that drive the effects documented elsewhere. The fact that the null result is robust to a variety of estimation designs and inference methods further supports the conclusion that the lack of effect is not due to methodological shortcomings, but rather reflects the substantive nature of the environmental provisions in China's PTAs during this period. Moreover, the findings are somehow informative for policy makers, as they suggest that the mere inclusion of environmental provisions in trade agreements may not be enough to achieve desired trade outcomes if those provisions are just loose cooperation language, even in the case of the largest exporter in the world. Instead, the design and content of the provisions themselves, including specificity and enforceability, are likely to be crucial for achieving meaningful trade effects.

The reminder of the paper is as follows. Section 2 reviews the related literature. Section 3 describes the data and the agreements. Section 4 presents the empirical strategy. Section 5 reports the results, with robustness analysis integrated as subsections. Section 6 concludes.

2 Related Literature

A large body of evidence documents that preferential trade agreements increase trade between member countries. Baier and Bergstrand (2007), using a panel gravity model that accounts for the endogeneity of agreement formation, estimate that an FTA roughly doubles bilateral trade over a ten-year period, a finding substantially larger than earlier cross-sectional estimates that did not control for selection into agreements. Subsequent work has refined this result by showing that the overall trade effect depends on agreement characteristics. Baier and Bergstrand (2009) use matching econometrics to estimate the causal effect of FTAs on trade and find that the magnitude varies with the depth and enforceability of the agreement, rather than being a uniform response to preferential market access. Medvedev (2010) confirms that the trade effects of PTAs are heterogeneous along several dimensions, including the type of members involved and the timing of liberalization. More broadly, Freund and Ornelas (2010) survey the literature on regional trade agreements and conclude that the distinction between shallow agreements, which primarily reduce tariffs, and deep agreements, which extend to regulatory harmonization, investment, services, and other behind-the-border provisions, is central to understanding their economic effects.

This distinction between shallow and deep agreements has been formalized through provision-level coding efforts. Hofmann, Osnago, and Ruta (2017) built the World Bank's Deep Trade Agreements database, which classifies the content of over 280 agreements into dozens of policy areas, and showed that agreements with similar headline coverage can differ dramatically in substantive provisions. Dür, Baccini, and Elsig (2014) constructed the DESTA depth index as an independent coding across seven policy areas, confirming the same heterogeneity from a different conceptual framework. Using these data, Mattoo, Mulabdic, and Ruta (2022) revisited the classical Vinerian question of trade creation and trade diversion, finding that deep agreements generate more trade creation and less trade diversion than shallow ones, and that some deep provisions have a public-good character that increases trade even with non-members. Neri and Santacreu-Vasut (2023) extend this analysis to the firm level, showing that deeper agreements benefit exporters unevenly across sectors, with industries more exposed to non-tariff barriers gaining disproportionately from provisions that address regulatory cooperation. Baccini, Pinto, and Weymouth (2017) provide further evidence of heterogeneity at the firm level, documenting that the distributional consequences of preferential trade liberalization are concentrated among larger and more productive firms. Taken together, these results point to a consistent conclusion: the effects of trade agreements depend critically on *what* is inside them. This raises a natural question: what about environmental clauses?

The interaction between environmental regulation and trade has been the subject of a large theoretical and empirical literature, organized around two competing mechanisms. The first is the pollution haven hypothesis, formalized by Copeland and Taylor (1994), which predicts that stricter environmental regulations increase compliance costs for pollution-intensive producers, causing production to shift toward countries with weaker standards. The mechanism operates through the composition channel: as regulation tightens in one country, its comparative advantage in dirty goods erodes, and trade patterns adjust accordingly. The second mechanism is the green upgrading or Porter hypothesis (Porter and Linde, 1995), which predicts the opposite: well-designed environmental regulations can trigger innovation and improve firm competitiveness, ultimately more than offsetting the direct costs of compliance. Jaffe and Palmer (1997) provide early empirical evidence on this channel, using a panel of US manufacturing industries to show that regulatory stringency is associated with increased patenting activity, although they find limited evidence that the resulting innovations translate into measurable competitive gains. These two mechanisms are not mutually exclusive, and whether environmental provisions in trade agreements push trade patterns in one direction or the other is ultimately an empirical question.

The empirical literature on the pollution haven hypothesis has produced mixed results, in large part because of the difficulty of separating the effect of environmental regulation from other determinants of trade patterns. Copeland and Taylor (2004) provide an extensive survey, concluding that while the theoretical logic is compelling, the evidence is sensitive to how one controls for confounding factors. Antweiler, Copeland, and Taylor (2001) decompose the effect of trade on pollution into scale, technique, and composition channels using data on sulfur dioxide concentrations across 43 countries over the period 1971–1996, and find that the technique effect dominates: freer trade appears, on net, to reduce pollution, because the income gains from openness induce stricter regulation. Brunnermeier and Levinson (2004) review the evidence on environmental regulation and industry location and reach a cautious conclusion: while there is some evidence that regulatory differences affect the location of pollution-intensive production, the effects are small relative to other determinants of industrial geography. Levinson and Taylor (2008) take on this challenge directly by instrumenting for abatement costs. Using trade data for 130 US manufacturing industries over the period 1977–1986, they find that industries with the largest increases in pollution abatement costs experienced the largest increases in net imports, consistent with the pollution haven prediction once one accounts for the econometric pitfalls that plagued earlier work. The dirty-industry classification used in the present paper draws on this tradition, specifically the taxonomy introduced by Mani and Wheeler (1998) and Low and Yeats (1992), who identified a set of pollution-intensive sectors — pulp and paper, industrial chemicals, petroleum refining, iron and steel, non-ferrous metals — that have become the standard reference for empirical work on trade and pollution.

Whether environmental provisions embedded in trade agreements can produce effects analogous to those of unilateral environmental regulation has been addressed by a small but growing literature. In a recent systematic review, Gutsch et al. (2024) survey the evidence on the effects of environmental provisions in trade agreements on businesses and economies, concluding that the literature remains thin and that the results are highly dependent on the type of provision and the empirical setting. Felbermayr, Peterson, and Wanner (2025) reach a similar conclusion in their survey of trade and environmental policies, noting that the interaction between trade policy and environmental policy is theoretically ambiguous and empirically context-dependent. Against this backdrop, a handful of empirical studies have produced results that point in a consistent direction.

Brandi et al. (2020) is the most direct antecedent of this paper. Combining the TREND environmental provision database (Morin, Pauwelyn, and Hollway, 2018) with bilateral trade flows from UN Comtrade for a panel of over 180 countries, they estimate the effect of different types of environmental provisions on the composition of developing countries' exports. Their key finding is that the effect depends critically on the type of provision: trade-restrictive provisions, which impose conditions on market access, reduce dirty exports by about 5 percent of the mean; liberal provisions, which facilitate trade in environmental goods, raise the green export share. Provisions couched in general cooperation language, by contrast, show no measurable effect. The estimation relies on a gravity framework at the aggregate bilateral level, with country-pair and year fixed effects, and the identification comes from cross-sectional variation in the type and number of environmental provisions across roughly

680 agreements. Berger et al. (2020) use a similar gravity framework and find that membership in PTAs with more environmental provisions is associated with lower bilateral exports overall, with the effect concentrated in South-North trade flows and not robust to PPML estimation. The contrast between these two findings is itself instructive: Brandi et al. (2020)'s result survives because their decomposition isolates the specific provisions that carry a trade mechanism, while Berger et al. (2020)'s aggregate index conflates provisions with and without such mechanisms.

Abman, Lundberg, and Ruta (2024) exploit a different margin of variation. Rather than comparing countries with and without environmental provisions, they compare agreements that contain specific clauses on forest preservation and biodiversity protection with agreements that do not, using satellite-based deforestation data as the outcome. Their identification strategy relies on within-agreement variation — not all agreements with environmental chapters include forest-specific clauses — and they find that these clauses offset the increase in deforestation that typically follows the entry into force of a trade agreement. The finding is important because it demonstrates that environmental provisions can have real environmental effects, but only when they contain a specific and identifiable mechanism. Baghdadi, Martinez-Zarzoso, and Zitouna (2013) provide additional evidence along similar lines, showing that PTA partners with environmental provisions experience convergence in emissions of sulfur dioxide and carbon dioxide, using a gravity-type model estimated with PPML on a panel of bilateral trade relationships over the period 1980–2005.

On the methodological side, the closest precedent is Cherniwchan (2017), who uses within-plant US manufacturing data around the implementation of NAFTA to identify the effect of tariff reductions on emissions intensities. His finding that emissions changes operate through individual producers' behavior rather than through market selection demonstrates the value of micro-level data for disentangling the channels through which trade policy affects environmental outcomes. However, his result concerns a *level* effect of trade liberalization on emissions — the kind of effect that the present design cannot identify, because environmental provision depth enters in force with the agreement and is absorbed by firm-destination-year fixed effects (Section 4). Shapiro (2021) documents that most countries' tariff structures implicitly subsidize pollution-intensive sectors and tax clean ones, a systematic environmental bias that environmental chapters are meant to work against. His replication data, which include China-specific CO_2 emission intensities by industry, supply the continuous pollution-intensity measure used as a robustness check in Section 5.

A recent strand of work has brought the question specifically to China. Yue and Lin (2024) study the effect of environmental provisions in PTAs on the export of energy-consuming products from China, finding that provisions inhibit these exports. Zhu and C. Sun (2026) use the

TREND database to examine whether environmental provisions in China's trade agreements promote green transformation, and report that EPs reduce the export of dirty goods while increasing the export of green products. Zhu and C. Sun (2025) focus on a different channel, documenting that environmental provisions in PTAs are associated with improvements in the quality of Chinese firms' exports and with increased imports of intermediate goods, consistent with a green upgrading mechanism. These studies provide suggestive evidence that environmental provisions may affect Chinese trade patterns, but they share two important limitations that the present paper addresses. First, none of them employs a staggered difference-in-differences design that accounts for the heterogeneous timing of agreement entry into force, which is a concern given the well-documented biases that arise when treatment turns on at different times across units (Callaway and Sant'Anna, 2021; Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021). Second, the environmental provision indices used in some of these studies vary with agreement characteristics that are themselves endogenous to the bilateral relationship, raising concerns about reverse causality. The present paper addresses the first concern through a triple-difference design with firm-destination-year fixed effects that absorbs the agreement itself and isolates the differential response of green and dirty products, and addresses the second by using two independent codings of provision content — the World Bank Deep Trade Agreements database and TREND — that are constructed from the text of the agreements themselves rather than from observed trade outcomes.

What emerges from this body of work is a consistent pattern: environmental provisions in trade agreements can have measurable effects on trade composition and environmental outcomes, but only when they contain specific, enforceable clauses with an identifiable mechanism. General cooperation language, which characterizes the vast majority of environmental chapters, does not produce detectable effects in studies that decompose provisions by type. This paper occupies the mirror position of Brandi et al. (2020): while they identify effects driven by the subset of agreements with specific provisions, the Chinese agreements in the period under study are dominated by exactly the kind of broad cooperation language that produced no effect in their decomposition. The present null result is therefore not a puzzle, but rather the predicted outcome of the existing literature's own heterogeneity findings, applied to a sample where the treatment lacks the specificity and enforceability that drive the effects documented elsewhere. What this paper adds is the granularity of the test: rather than aggregate bilateral flows, it exploits the universe of Chinese customs transactions to ask whether the same firm shifted its product mix in response to environmental provisions, a question that aggregate data cannot answer. It also adds a methodological contribution, engaging the growing literature on inference with few treated clusters (Abadie et al., 2023;

Cameron, Gelbach, and Miller, 2008; Conley and Taber, 2011; MacKinnon and Webb, 2017) to show that a result that appears significant under asymptotic inference dissolves under bootstrap and permutation tests — a cautionary finding for any study in which the treatment varies across a small number of policy units.

3 Data

3.1 Chinese PTAs and their environmental content

Between 2000 and 2015, China's preferential agreements in force cover 25 destination economies (Table 1). Three features of this treatment landscape shape everything that follows.

Table 1: Treatment map: Chinese PTAs in force, 2000–2015

Agreement (partners)	In force	Notes
Bangkok Agr./APTA (BGD, IND, KOR, LAO, LKA)	2002/2005	minimal EP content
ASEAN–China (10 members + Timor-Leste, see note)	2005	one agreement, 11 destinations
Chile	2006	
Pakistan	2007	
New Zealand	2008	first substantive environmental chapter
Singapore	2009	also ASEAN member
Peru	2010	
Costa Rica	2011	
Iceland; Switzerland	2014	richer environmental content
Australia; South Korea	2015	Korea: one of three within-depth switchers
Hong Kong; Macao (CEPA)	2003	excluded: entrepôt, sui generis

Entry-into-force years from the World Bank Deep Trade Agreements database and TREND. Within-depth switchers ar destinations whose EP depth increases after initial entry: Laos (Bangkok Agreement 2002 → ASEAN 2005), Singapor (ASEAN 2005 → bilateral FTA 2009), and South Korea (Bangkok Agreement 2002 → 2015 FTA). Timor-Leste is coded a an ASEAN–China party in the source databases; it accounts for 0.02% of observations and main coefficients are unchange beyond the fourth decimal if it is excluded.

First, the ASEAN agreement covers eleven destinations with identical values of every EP index: the effective number of independent agreements is roughly 14, not 25. Treating these as separate variation would be misleading; they constitute a single treatment in all but name.

Second, EP content is fixed at signature and almost never changes afterwards. Only three destinations (Laos, Singapore, and South Korea) exhibit within-country variation over time, and in each case depth rises, never falls. There is no treatment reversal in the sample, which means estimators designed for treatments that switch on and off are not needed here and would be misspecified.

Third, Hong Kong and Macao (CEPA, 2003) are entrepôt economies. They account for

24.4% of treated observations and 50.1% of treated export value, but a large fraction of what the customs data record as Chinese exports there are goods transshipped and re-exported to third countries at a markup (Feenstra and Hanson, 2004). Including them would conflate demand from the true final destination with the bilateral fixed effects meant to absorb it. Both economies are excluded from the main specification; results with their inclusion are reported in Section 5.

Table 2 shows that entry into the treated group is concentrated in two large cohorts (2002 and the 2005 ASEAN accession) and then trickles in one destination at a time in most subsequent years. The smallest cohorts contain a single economy (Chile 2006, Pakistan 2007, and so on through Australia 2015). Of the 236 destinations in the panel, 23 are ever treated and 213 are never treated within the sample window.

Table 2: Treatment-timing cohorts

Entry year	N destinations	Destinations
2002	5	Bangladesh, India, Korea, Laos, Sri Lanka
2005	10	Brunei, Cambodia, Timor-Leste, Indonesia, Malaysia, Myanmar, Philippines, Singapore, Thailand, Vietnam
2006	1	Chile
2007	1	Pakistan
2008	1	New Zealand
2010	1	Peru
2011	1	Costa Rica
2014	2	Iceland, Switzerland
2015	1	Australia
Total treated	23	
Never treated	213	

Excludes Hong Kong and Macao (2003; see text). South Korea's 2015 upgrade to the bilateral FTA is a within-country depth increase, not a new cohort entry. Source: `New/Output/Diagnostics/B_treatment_entry.csv`.

EP content is measured with two independent codings. The World Bank Deep Trade Agreements database (Hofmann, Osnago, and Ruta, 2017) yields `WB_EP_Depth`, the count of environmental-law provisions in force with destination d at time t, and a non-environmental `TotalDepth` control. TREND (Morin, Pauwelyn, and Hollway, 2018) yields `TREND_EP_Count` and sub-indices covering green market access, enforcement stringency, and hard versus soft provisions. Using two codings independently constructed from different frameworks guards against the idiosyncrasies of either; as shown below, they agree on the main findings.

3.2 Customs data and product classification

The trade data are the universe of Chinese export customs transactions for 2000–2015, aggregated to the firm–HS6–destination–year level: 49.2 million observations (45.8 million once Hong Kong and Macao are excluded), roughly 462,000 exporting firms, around 5,000 HS6 products, and 236 destinations. The outcome is the log of export value; no trimming or winsorization is applied in the main specification, with the design-based inference battery serving as the primary guard against influential observations (trimming is applied as a robustness check in Section 5).

Green products are the 248 HS6 codes of the OECD Combined List of Environmental Goods (CLEG), as compiled in Sauvage (2014). The customs panel uses the HS1996 vintage while the OECD list is native HS2012, so the list was translated: 246 of the 248 codes have a unique concordance and are mapped directly; the two that do not (codes 871411 and 871419, a single HS heading split into two sub-codes) are retained at their original codes and flagged. Export-value continuity checks around the 2007 vintage revision find no suspicious breaks on any translated code.[1]

Dirty products belong to the classic pollution-intensive sectors of Mani and Wheeler (1998) and Low and Yeats (1992): pulp and paper, industrial chemicals, petroleum refining, iron and steel, non-ferrous metals, and cement (in an extended list). These are mapped to HS1996 via the WITS/UNSD HS1996–ISIC Rev.3 concordance, yielding 1,139 codes. Seventeen codes that appear in both lists (materials produced by dirty industries but used for environmental purposes, such as some insulation products) are assigned to the green list; the two categories are mutually exclusive.

Green products account for 11.5% of firm-level observations and dirty products for 7.0%. The remaining neutral products form the comparison group. Under any agreement with EPs, 20.3% of observations are exposed.

3.3 Estimation samples

The analysis rests on four estimation objects built from the same underlying panel (Table 3).

The *full firm-level panel* is the main object. The *collapsed panel* aggregates to HS6–destination–year cells (3.77 million; outcome: within-cell mean of log exports; regressions

[1] As a check on the classification itself, the specification was re-estimated restricting green products to the 54 codes of the APEC Environmental Goods List (Sauvage, 2014), on which there is explicit multilateral political consensus and which form a complete subset of the OECD list. With the WB index the green coefficient flips sign to +0.0050 (s.e. 0.0127, $p = 0.69$); with TREND it moves from +0.0018 to +0.0032 ($p = 0.13$). Both remain indistinguishable from zero, with standard errors roughly double, as expected from restricting to 22% of the original green sample. The null is not driven by borderline products in the broader list.

Table 3: Descriptive statistics

Firm-level panel (excl. HK–MO)	
Observations (firm–HS6–dest–year)	45,781,211
Exporting firms	≈462,000
HS6 products	≈5,000
Destinations	236
of which treated (EP depth > 0)	23
Effective agreements	≈14
Share of observations: green products	11.5%
Share of observations: dirty products	7.0%
Share of observations under an EP agreement	20.3%
Collapsed panel (HS6–dest–year)	
Cells	3,773,498
of which green / dirty	8.4% / 14.0%
Estimation sample (post-singleton)	3,681,023
PPML zero-filled grid (HS6–dest–year, excl. HK–MO)	
Cells	8,179,904

Hong Kong and Macao excluded throughout (they account for 24.4% of treated observations and 50.1% of treated export value). Product classification as described in the text.

weighted by cell counts) and serves both as a computational workhorse for the bootstrap and permutation battery and as an internal cross-check. The *zero-filled PPML grid* (8.2 million cells) adds product–destination combinations with no recorded trade and is used for extensive-margin estimates. The *within-firm share panel* (13.3 million firm–destination–year observations) recasts the outcome as each firm's green export share toward a destination; it is reported as descriptive evidence only, for the identification reason given in Section 4.

Because the triple-difference compares green and dirty products to neutral ones, the credibility of the comparison group can be probed directly. Four control-group subsamples, each tightening the comparison in the direction of a specific threat, are described in Table 4.

The reading rule that applies throughout: a genuine composition effect survives every tightening of the comparison group; an artifact of the comparison moves when the comparison moves.

Table 4: Control-group subsamples

Subsample	Threat addressed
C-prod-HS4 neutral products in the same HS4 family as a green code	Neutral products differ from green/dirty
C-overlap products exported both to treated and untreated destinations	Extrapolation outside common support
CEM destinations destinations matched on year-2000 GDP p.c., growth, MFN tariff	Selection of partner countries
Deep vs. shallow PTA partners only, split at median EP depth	The “any-agreement” contrast itself

C-prod-HS4 (106 HS4 families; 20.5% of observations) should be read alongside C-overlap rather than alone: multi-produ may reallocate toward core products when trade costs fall (Eckel and Neary, 2010), displacing neutral products in the sa family and biasing the contrast. C-overlap is immune by construction (98.5% of HS6 products are exported both to and never-treated destinations). The deep-vs-shallow design drops untreated destinations entirely (16 deep, 9 shallow countries); the few-cluster inference battery of Section 4 handles the small number of shallow clusters. CEM matching: 16 + 40 control destinations.

4 Empirical Strategy

4.1 Why a level effect is not identifiable

Regressing log exports on EP depth with fixed effects — the natural first move — is uninformative in this setting. Documenting this as a diagnostic rather than suppressing it matters: the failure has a specific structure that is itself informative.

EP depth varies at the destination–year level, enters in force with the agreement, and afterwards almost never changes (only three destinations have within-country variation, and depth never falls). With roughly 14 agreements and near-zero within-destination variation, “environmental depth” and “having a deep agreement” are indistinguishable. The saturation ladder makes the point empirically: moving from sparse fixed-effects structures (firm–product–destination and year) to saturated ones (adding product–year and pair-level interactive effects), the EP-depth coefficient falls monotonically from small positive and nominally significant to an exact, precisely estimated zero. The pattern — significance in under-saturated specifications with understated standard errors — is the classic signature of selection into agreements rather than a causal effect (Bertrand, Duflo, and Mullainathan, 2004; Goodman-Bacon, 2021).

One uncomfortable consequence follows directly: this design cannot supply evidence of first-stage “bite” — a demonstration that signing an EP-heavy agreement shifted some intermediate outcome before composition is examined — because that level effect is exactly what is confounded with the agreement itself. The saturation ladder is the closest substitute available. It shows that, whatever the agreement did at the level of trade, EP depth is not separately identified there. The paper's evidence is accordingly built entirely on the composition margin.

4.2 The triple-difference specification

The target is an average treatment effect on the treated: the effect of one additional unit of environmental depth on the gap between green (or dirty) and neutral log exports, within firm–destination–year cells that experience treatment. In potential-outcomes terms: $\beta_1 = E[(y_{fpgdt}(1) - y_{fpgdt}(0)) - (y_{fpndt}(1) - y_{fpndt}(0)) \mid d, t \text{ treated}]$, where $y(1)$ and $y(0)$ denote log exports of green product g versus neutral product n under deeper and unchanged environmental content respectively, holding the agreement fixed. This is a per-cell (per firm–destination–year) average, not a per-firm average. The main specification is

$$\ln x_{fpdt} = \beta_1\, EP_{dt}{\times}g_p + \beta_2\, EP_{dt}{\times}b_p + \gamma_1\, TD_{dt}{\times}g_p + \gamma_2\, TD_{dt}{\times}b_p + \theta_{fpd} + \theta_{fdt} + \theta_{pt} + \varepsilon_{fpdt}, \quad (1)$$

where f indexes firms, p HS6 products, d destinations, and t years; g_p and b_p are green and dirty indicators; EP_{dt} is environmental depth and TD_{dt} non-environmental agreement depth.

The firm–destination–year fixed effects θ_{fdt} are the load-bearing element. They place a separate intercept on every cell, so that anything constant within a firm's relationship with a market in a year — the agreement dummy, overall agreement depth, destination demand, the non-random selection of destinations into agreements — drops out exactly. Identification comes from comparing green and dirty products to neutral ones *within* the same cell as EP content varies across destinations and over time. Mechanically, selection into agreements can contaminate β_1 only if it operates *differentially* on green versus neutral products inside the same cell — the residual threat that the destination-trend exercise addresses. The fixed effect θ_{pt} absorbs global product-level shocks such as the world solar market expansion, and θ_{fpd} absorbs firm-level specialization patterns.

The omitted category is neutral products. This means $EP \times$ *green* is *already* the identifying contrast (green versus neutral); the green–dirty differential is not the parameter of the design and should not be read as one. It is a useful diagnostic, but it is not what the regression estimates.

The TD interactions separate environmental from general agreement content. `TotalDepth` counts all non-environmental provisions rather than isolating those most relevant to green trade, so it is an imperfect proxy. An imperfectly measured control does not attenuate the coefficient of interest in a signable direction: it under-controls, and β_1 absorbs whatever share of general-depth variation is correlated with the omitted part. Rather than assert a direction, Section 5 reports the green coefficient under four alternative depth controls and shows that the point estimate moves within a band narrower than one standard error (Table 9). The collinearity deserves to be stated plainly: across the 223 treated destination–years in the estimation sample, WB EP depth and non-environmental TotalDepth correlate at 0.91 raw and 0.96 after destination and year demeaning; the raw VIF is 5.8, somewhat above the maximum of 4.6 that Brandi et al. (2020) report for their provision-type variables. The TREND count is less collinear with TotalDepth (0.50 raw, 0.85 within), which is one reason the WB confidence intervals are wider; that two codings with such different degrees of overlap return the same null is reassuring, since a result driven by the inability to separate environmental from general depth would be expected to differ between them. Standard errors are clustered by destination, the level of treatment variation (Abadie et al., 2023).

A further identification concern works in a conservative direction. If deeper environmental chapters tend to be signed where green demand is already rising, EP_{dt} is positively correlated with an omitted destination-specific trend, which would bias β_1 toward a spurious positive. Because β_1 is an imprecise zero, this channel, if present, implies the true effect is zero or negative — the null is a conservative rather than a lenient reading of the data.

One qualification about the estimator, not the target. Environmental depth is a *continuous* dose switching on at *staggered* dates. Under that combination the TWFE coefficient is not in general the ATT: it is a weighted average of dose- and cohort-specific effects whose weights need not be convex, and dose comparisons face selection into dose that parallel trends alone does not rule out (Callaway, Goodman-Bacon, and Sant'Anna, 2024; Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021). Two features limit what this costs in this particular setting. Estimating a null makes the weighting problem less consequential, since a weighted average of near-zero effects stays near zero unless the underlying effects are large and of opposite signs, of which the sub-index and leave-one-out evidence gives no indication. And the permutation test is entirely agnostic about weighting: it asks whether the observed coefficient — whatever weighted average it represents — is unusual relative to the same statistic under randomly reassigned EP profiles. β_1 is therefore best read as a weighted average of composition responses across doses and cohorts. A continuous-dose estimator in the spirit of Callaway, Goodman-Bacon, and Sant'Anna (2024) is the natural next step for this design.

4.3 The two panel implementations

The full firm-level panel is estimated using iterative singleton removal (Correia, Guimarães, and Zylkin, 2021), which eliminates 24.3 million observations that contribute no identifying variation. The surviving sample of 21.5 million observations retains 47% of all observations but 70% of aggregate export value (the removed observations are predominantly one-off, small flows), and the product composition is essentially unchanged: green products move from 11.5% to 11.9% of observations, dirty from 7.0% to 6.4%. Within the estimation sample the composition contrast is identified in treated firm–destination–year cells that contain both a green (or dirty) and a neutral product: 26% of treated cells for the green margin and 12% for the dirty, with a median of one green product per identifying cell and three products per treated cell overall.

The collapsed panel aggregates the firm dimension to HS6–destination–year cells (3.77 million; outcome: within-cell mean of log exports; weights: number of underlying observations; fixed effects $pd + dt + pt$). The weights are not a modelling choice: they are the condition under which the weighted cell-level regression is algebraically identical to the unweighted micro regression under the same fixed effects. This equivalence is verified to seven significant figures: -0.0045685 versus -0.0045685 on the green margin; -0.011873 versus -0.011873 on the dirty margin.

What the collapsed panel does not reproduce is the firm dimension, and the two implementations should be compared with that distinction in mind. On the green margin they agree substantively (-0.0046 collapsed, -0.0023 full panel; both indistinguishable from zero). On the dirty margin they differ by a factor of roughly 2.7 (-0.0119 versus -0.0044), and the gap is a result rather than a discrepancy: the collapsed panel compares products across the firms serving a given destination–year, so any shift in which firms export dirty goods enters the coefficient; θ_{fdt} removes that channel and leaves only reallocation within the firm. About three-fifths of the collapsed dirty coefficient reflects between-firm composition rather than within-firm response. This is a reason the full panel is the reference specification and the collapsed panel the workhorse for the inference battery.

4.4 Inference with few treated clusters

There are 225 destination clusters in the full-panel specification (236 in the collapsed panel) but only 23 treated ones, and effectively 14 agreements.[2] Asymptotic cluster-robust inference is unreliable in this regime, so every headline estimate is subjected to three tests.

Wild cluster bootstrap (B = 9,999 replications, Cameron, Gelbach, and Miller, 2008;

[2] Every estimate was produced twice: in STATA (`reghdfe`, `boottest`, `ppmlhdfe`, `eventstudyinteract`) and in R (`fixest`, `fwildclusterboot`). Point estimates agree to at least eight significant digits throughout. The numbers reported are the STATA ones, as the do-files are the more direct route to replication. Two quantities legitimately differ between implementations and are discussed where they arise: bootstrap and permutation p-values, which carry Monte Carlo error, and the Sun–Abraham standard errors of Appendix C, which reflect a genuine methodological difference between `eventstudyinteract` and `fixest::sunab` (see Appendix C).

Roodman et al., 2019) is computed on Frisch–Waugh residualized data — fixed effects demeaned once, not re-estimated at each draw. Two approximations apply in the collapsed-panel implementation only. The bootstrap treats pt as if nested within the destination cluster (it is not; pd and dt are), and the small-sample corrections see the four residualized regressors rather than the four plus absorbed fixed effects. Neither approximation applies to the full-panel bootstrap, which runs natively after `reghdfe` and is the version quoted for the headline intervals.

Permutation inference (Fisher, 1935): entire EP profiles (depth and timing) are randomly reassigned 1,000 times across treated destinations, asking whether the observed coefficient stands out from placebos. Three features should be stated explicitly. The null is the restricted one — which treated destination holds which profile — not treated versus never-treated. Depth and timing are permuted jointly, so the test cannot separate environmental content from the date of entry. The effective support is far smaller than 1,000: the eleven ASEAN destinations carry identical profiles, leaving roughly nine distinct profiles; the permutation distribution is correspondingly granular, which is why the R and Stata implementations give $p = 0.235$ and $p = 0.278$ respectively for the dirty margin — both correct, and both far above any conventional threshold.

5 Results

5.1 The green margin: a bounded null

Table 5 reports the main estimates. In the full firm-level panel, the EP×green coefficient is -0.0022 (s.e. 0.0039) with the WB index and -0.0001 (s.e. 0.0010) with TREND. The joint F-test on all four composition interactions is $p = 0.31$ (WB) and $p = 0.71$ (TREND). In the collapsed panel the WB estimate is -0.0046; wild cluster bootstrap $p = 0.65$; permutation $p = 0.60$. The observed green coefficient sits comfortably in the middle of the placebo distribution.

To calibrate magnitudes, consider the closest benchmark. Brandi et al. (2020) find that one liberal environmental provision raises the green share of developing-country exports by around 0.4 percentage points — roughly $+0.16$ log points in the metric of equation (1) — per provision. Section 4.4 argued that asymptotic inference is unreliable with 23 treated clusters; that argument extends to interval estimates, so the bounds cited here are wild cluster bootstrap intervals. The full-panel 95% bootstrap confidence interval for EP×green is $[-0.0353, +0.0355]$ per WB provision (collapsed: $[-0.0182, +0.0317]$). The corresponding asymptotic interval, $[-0.0100, +0.0055]$, is roughly six times narrower — reporting that level

Table 5: Triple-difference estimates: EP content and export composition

	WB EP depth		TREND EP count	
	(1) ×green	(2) ×dirty	(3) ×green	(4) ×dirty
Panel A. Full firm-level panel				
EP interaction	−0.0023	−0.0044	−0.0001	−0.0009
	(0.0039)	(0.0022)	(0.0010)	(0.0006)
Asymptotic p	0.57	0.052	0.92	0.15
Bootstrap p	0.69	0.18	0.93	0.18
Firm–product–dest FE	Yes	Yes	Yes	Yes
Firm–dest–year FE	Yes	Yes	Yes	Yes
Product–year FE	Yes	Yes	Yes	Yes
Observations	21,519,511			
Clusters	225 destinations			
Joint F (p-value)	0.31		0.71	
Panel B. Collapsed panel (HS6–dest–year cells)				
EP interaction	−0.0046	-0.0119^{***}	+0.0018	+0.0004
	(0.0070)	(0.0030)	(0.0018)	(0.0016)
Asymptotic p	0.51	<0.001	0.32	0.83
Bootstrap p	0.65	0.07	0.39	0.86
Permutation p	0.60	0.28	0.16	0.82
Product–dest FE	Yes	Yes	Yes	Yes
Dest–year FE	Yes	Yes	Yes	Yes
Product–year FE	Yes	Yes	Yes	Yes
Cells	3,681,023			
Clusters	228 destinations			

All specifications include TotalDepth×green and TotalDepth×dirty controls (not shown). Wild cluster bootstrap: B = 9,999 (Cameron, Gelbach, and Miller, 2008; Roodman et al., 2019). Permutation: 1,000 reassignments of entire EP profiles across treated destinations. Bootstrap 95% CIs — Panel A (1)–(4): $[-0.035, +0.036]$, $[-0.043, +0.011]$, $[-0.0037, +0.0025]$, $[-0.0027, +0.0007]$; Panel B: $[-0.018, +0.032]$, $[-0.018, +0.002]$, $[-0.0017, +0.0071]$, $[-0.0031, +0.0056]$. $^{***}p < 0.001$ (asymptotic only; does not survive bootstrap or permutation — see Section 5.4).

of precision would claim the kind of certainty this paper's own inference section declares unwarranted.

On the bootstrap bound, the upper limit is about a quarter of the Brandi equivalent effect: the design rules out their point estimate and anything above roughly a quarter of it, and is uninformative below that threshold. This is a weaker statement than asymptotic inference would support, and it is the one the design can sustain. In per-standard-deviation terms, a one-SD increase in WB EP depth (approximately 2.383 provisions, inclusive of never-treated destination–years at EP = 0) moves green relative to neutral exports by at most −8.4% to +8.5% at the bootstrap bounds.

Table 6: Benchmark against Brandi et al. (2020): equivalent effect and comparison with the confidence intervals

Margin	Brandi effect (log points)	Asy. CI upper	WCB CI upper/coef.	Ratio
Green (liberal)	0.1570	0.0055	0.0355	1/4 (WCB)
Dirty (restrictive)	-0.0513	-0.0044 (point est.)	–	1/12

5.2 Stability across control groups

Across nine designs — the four control-group subsamples, the two baseline specifications, and three full-panel variations — the EP×green coefficient stays between −0.0009 and −0.0046 and is never significant (Table 7). The stability is the result: an artifact of the comparison group would move when the comparison group changes.

5.3 Dynamics

Figure 1 plots the event study of the green and dirty versus neutral gap around entry into force, using never-treated destinations as controls. Differential pre-trends are flat for both margins. Before taking that as reassurance, a clarification is in order: parallel trends is a claim about the unobserved counterfactual path of treated destinations *after* treatment. A flat pre-trend rules out the most easily detectable violations — a composition gap already in motion before the agreement — but does not establish that the assumption holds exactly thereafter. The exercise is a falsification, not a proof.

A mild negative drift for green products in the late post-period (bins $\geq +5$) is identified only by early cohorts (ASEAN 2005, Chile 2006, Pakistan 2007, New Zealand 2008). It disappears under the Sun–Abraham estimator applied to the destination-level composition gap: the aggregated ATT is -0.042 ($p = 0.27$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap — consistent with early-cohort heterogeneity rather than any effect of treatment

Table 7: Stability of the EP×green coefficient across estimation designs (WB index)

Sample / design	Obs.	EP×green	Asymptotic p
Full panel (firm level)	21.5M	−0.0023	0.57
Collapsed panel	3.7M cells	−0.0046	0.51
Control-group subsamples			
C-prod-HS4 (within-HS4 neutral products)	3.8M	−0.0009	0.86
CEM-matched destinations	13.7M	−0.0023	0.54
Deep vs. shallow (PTA partners only)	5.3M	−0.0022	0.53
Common support (C-overlap)	21.5M	−0.0022	0.57
Full-panel variations			
With controls	19.3M	−0.0002	0.94
Excluding ASEAN	19.2M	−0.0025	0.48
Including HK–Macao	23.6M	−0.0009	0.80
Fixed effects as in baseline		Yes (all rows)	
WB index		Yes (all rows)	

Each row estimates equation (1) on the indicated subsample or design; asymptotic destination-clustered p-values. Fixed effects: $fpd + fdt + pt$ (full panel); $pd + dt + pt$ (collapsed). CEM matches on log GDP p.c., growth, and MFN tariff (year 2000). Deep/shallow split at the median of maximum EP depth: 16 deep vs. 9 shallow partner destinations. TREND analogues (where estimated): full panel −0.0001 ($p = 0.91$), common support −0.0001 ($p = 0.91$), deep vs. shallow −0.0004 ($p = 0.72$). Controls: MFN tariff, HHI, antidumping indicator.

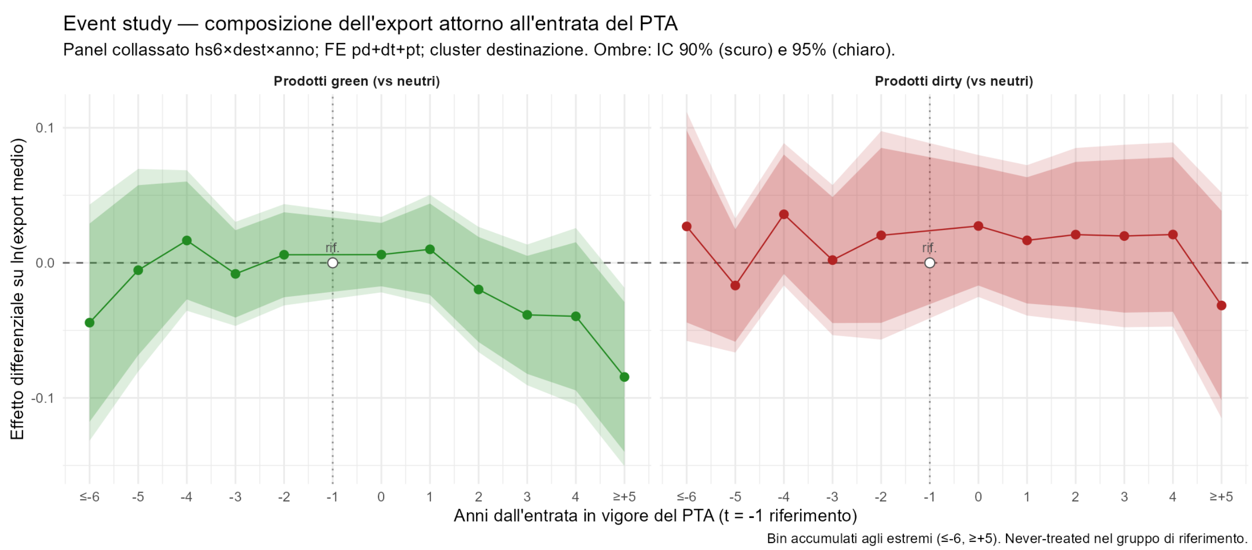

Figure 1: Event study: green and dirty vs. neutral products around PTA entry into force. Collapsed panel; FE $pd + dt + pt$; destination-clustered 95% confidence bands; $t = -1$ is the reference period; endpoint bins accumulate.

(L. Sun and Abraham, 2021). The Sun–Abraham estimator answers a narrower question than equation (1) — it binarizes treatment (EP present or absent, discarding depth), carries no depth control, and operates on a destination–year composition gap collapsed across products — so it should not be read as a replication. It is a diagnostic for staggered-timing and cohort heterogeneity, and that is how it is used here.

The full interaction-weighted event study is reported in Appendix C. With the standard errors produced by `eventstudyinteract` — which, unlike `fixest::sunab`, accounts for the uncertainty in the estimated cohort shares used as weights — no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on either margin. Two or three nominally significant coefficients appear among 28 estimates, which is what sampling noise produces under exactly flat pre-trends with 23 treated clusters. The ATT is null across all three specification variants: $p = 0.28$ in the baseline, $p = 0.11$ in the $[-6, +5]$ window, $p = 0.30$ excluding the 2014–2015 cohorts.

5.4 The dirty margin: anatomy of a false positive

The WB×dirty coefficient in the collapsed panel (-0.0119, asymptotic $p < 0.001$) warrants a subsection of its own. It illustrates, step by step, how a coefficient that looks robust under standard inference falls apart under scrutiny.

The wild cluster bootstrap raises the p-value to 0.07. Permutation inference on the exact estimating equation goes further: reshuffling entire EP profiles across the 23 treated destinations, 27.7% of placebo draws produce a coefficient as large in absolute value, giving $p = 0.28$. A coarser permutation on a destination–year–product-type aggregate reverses the sign entirely ($+0.005$, $p = 0.49$). The aggregation removes the within-cell heterogeneity across firms and product varieties that the disaggregated coefficient exploits, so the two tests are not answering exactly the same question, but they reach the same verdict. Sign instability under aggregation is itself evidence of fragility.

A leave-one-out exercise across the 23 treated destinations reveals the mechanism, and it is not the one the phrase "driven by a single country" usually implies. The *point estimate* is remarkably stable: it sits between -0.0097 and -0.0133 in every one of the 23 leave-one-out samples. Dropping large treated destinations such as India or Pakistan moves it by 1 to 4% while leaving the standard error essentially unchanged. Those are not the pivotal observations.

What is not stable is the *precision*. Dropping Australia moves the coefficient by 13% (to -0.0103) but nearly triples its standard error, from 0.0030 to 0.0087: it is this, not the movement of the estimate, that carries the p-value to 0.24. Dropping South Korea doubles the standard error, giving -0.0097 with $p = 0.09$. Australia and South Korea are therefore not outliers exerting leverage on the estimate. They supply the variation that identifies it, and removing them leaves the design without enough information to speak — rather than with a different answer. The right description is that the coefficient is identified by a thin slice of variation concentrated in two destinations, not that those destinations pull the estimate in the wrong direction.

The exercise with the DESTA depth control makes the point sharper. With TotalDepth, Australia is the pivotal destination; with DESTA, Australia's exclusion leaves the coefficient at -0.0110 with $p = 0.001$, and it is South Korea whose removal triples the standard error and gives $p = 0.14$. Which destination the precision depends on is itself a function of a modelling choice that has nothing to do with environmental content. A genuine identified effect does not behave this way.

The TREND index never shows the effect (bootstrap $p = 0.86$). In the full panel the coefficient is -0.0044 (asymptotic $p = 0.052$); the wild cluster bootstrap raises it to $p = 0.18$, with a 95% interval of $[-0.043, +0.011]$. The marginal signal under asymptotics is erased by robust inference on both panels independently. The honest reading is that the dirty margin is a false positive of exactly the type few-treated-cluster designs can manufacture: an estimate that is overwhelming under asymptotic standard errors and that clears none of the same battery applied to the green margin. It belongs in the paper as a descriptive pattern and a methodological warning.

5.5 Provision bundling and the limits of heterogeneity analysis

Aggregate EP indices combine provisions with an identifiable trade mechanism and cooperation language that has none. On any aggregate null, this is the natural objection — and it is correct. The response here is not to refute it but to show how extreme the bundling is, and what that implies for interpretation.

Summing WB_EP_Depth across the 25 treated destinations gives 150 provisions checked in total; 5.3% fall in `GreenLiberalization` or `StandardsNonRegression`, the two sub-indices with a direct trade mechanism. TREND, an independently constructed and far more granular coding, reaches the same conclusion: of 437 provisions across the same destinations, `GreenMarketAccess` is 0.92%. At the agreement level, TREND's binding-obligations clause (`X5.01.01`) appears in exactly one of the 14–15 Chinese agreements in the database (China–Korea, 2015). Table 8 puts both codings side by side.

The structural consequence is stark. Across the 223 treated destination–years in the estimation sample, the two WB sub-indices with a direct mechanism are *perfectly* collinear ($\rho = 1.000$) and non-zero in only three country–years: Korea from 2015 and Switzerland from 2014, always in the same 1:3 proportion. Their interaction terms therefore load identically in

Table 8: Trade-mechanism content as a share of total checked provisions

	Total checked	Trade-mechanism provisions	Shar
WB (`GreenLiberalization` + `StandardsNonRegression`)	150	8	5.3%
TREND (`GreenMarketAccess`)	437	4	0.92%
	Agreements	With binding clause	
TREND (`X5.01.01` Binding obligations)	14–15	1 (China–Korea, 2015)	6.7%

"Total checked" sums the relevant depth index across the 25 treated destinations at each destination's maximum (post-entr value. Source: `Data/Merged/Merged_TREND_WB_Indices_Only.csv`.

the regression (identical t-statistics; coefficients in exact 3:1 ratio), and the VIF is unbounded. Contrast this with Brandi et al. (2020), who report a maximum VIF of 4.6 across provision-type variables on their 680-agreement panel: enough agreements with enough variation in provision type to separately identify the mechanism sub-components. With 14 agreements, this design cannot.

A reading rule applies to everything that follows: each sub-index replaces the aggregate index one at a time and is estimated on its own. The resulting coefficients are not partial effects holding other clause types constant; they absorb whatever variation in correlated sub-indices moves with each one. This is the only feasible approach, but it means the table is a set of separate comparisons, not an additive decomposition of the aggregate index.

What can be said: the mechanism-bearing bundle shows the same pattern as the aggregate. Green never responds (WB `GreenLiberalization`×green: $p = 0.90$; TREND `GreenMarketAccess`×green: $p = 0.38$). The dirty interaction is marginally negative on asymptotic p-values ($p = 0.04$–0.07), but this is the same asymptotics-only signal documented in Section 5.4, which does not survive robust inference.

The placebo sub-indices (without a trade mechanism) are more interesting and worth reporting carefully. TREND soft provisions show nothing on either margin (×green: $p = 0.27$; ×dirty: $p = 0.86$), consistent with the logic of the null. Regulatory space clauses do not: they carry +0.024 on green and +0.023 on dirty, and both survive the wild cluster bootstrap ($p = 0.046$ and $p = 0.022$, $B = 9{,}999$). This is the only place in the paper where a sub-index without a trade mechanism registers a signal that robust inference sustains.

Two features bound what can be read into it, and together they leave the conclusion weak. First, the two margins move in tandem: the green–dirty differential is +0.0017 with $p = 0.80$, meaning the coefficient describes green and dirty exports rising by the same amount against the neutral baseline — not a shift in composition toward cleaner goods. Second, and more fundamentally, the regulatory-space sub-index is not a clean placebo. It accounts for 71.5% of the TREND count across treated destination–years and correlates 0.90 with TotalDepth;

in this specification the depth control flips to significantly negative (×green: -0.0011, $p = 0.006$). Two regressors correlated at 0.90 splitting into a large positive and a large negative is the same bundling problem applied to the placebo, not a cleanly identified channel. The defensible conclusion is the weaker one: this design cannot separate a regulatory-space effect from general agreement depth, so neither the positive signal nor its absence can serve as a clean falsification test.

Enforcement provisions (dispute-settlement mechanisms attached to the environmental chapter) are null on both margins and both codings: WB ×green $p = 0.91$, ×dirty $p = 0.90$; TREND ×green $p = 0.78$, ×dirty $p = 0.71$.

This bundling result reconciles the null with Brandi et al. (2020). Their finding is not that environmental provisions generically matter; it is driven by the subset of agreements with explicitly liberalizing or restrictive clauses, identified separately because 680 agreements supply enough cross-agreement variation in provision type. The Chinese PTAs of this period are, on the same coding, overwhelmingly cooperation-only: mechanism-bearing content appears in two of the fourteen agreements and is perfectly collinear between its two components in-sample. A null on the aggregate index is what Brandi et al.'s own heterogeneity would predict for a sample of this composition. The two results are complementary. A second compounding difference concerns aggregation: Brandi et al. (2020) work with exporter–importer–year flows, where a composition effect can arise from the entry or exit of firms or products. The specification here saturates firm×destination×year fixed effects and identifies from reallocation *within* the same firm across products — a strictly more demanding test of the same hypothesis.

5.6 Extensive margin

Log-linear estimates condition on positive flows. If EPs create new green trade at the extensive margin, intensive estimates miss it (Santos Silva and Tenreyro, 2006). On a zero-filled HS6–destination–year grid (8.2 million cells), PPML estimates confirm the null: EP×green is $+0.0015$ ($p = 0.74$, WB) and $+0.0001$ ($p = 0.95$, TREND); EP×dirty is -0.030 ($p = 0.06$, WB) and $+0.003$ ($p = 0.55$, TREND). No green trade creation at the extensive margin; the dirty coefficient is at most marginally significant and consistent with the non-result reading of Section 5.4.

One limitation bears noting: this grid detects green products newly exported to a destination, but not a new firm beginning to export a product that other Chinese firms already ship there, which is a common form of firm-level entry. The extensive margin is probed, not exhausted.

5.7 Within-firm share: a descriptive complement

On the firm–destination–year panel of green export shares (13.3 million observations; FE: firm–destination and year; destination-clustered), the coefficient on WB EP depth is -0.0001 ($p = 0.50$) and on TREND -0.00006 ($p = 0.043$, asymptotic). Both are economically negligible: a one-standard-deviation increase in the TREND count moves the within-firm green share by three hundredths of a percentage point, and the marginal asymptotic significance carries the same few-cluster caveat as the rest.

This module is reported as a descriptive complement, not identifying evidence. The specification regresses the green share on the *level* of EP depth with firm–destination and year fixed effects — it does not include firm–destination–year absorption and therefore inherits the collinearity with the agreement that Section 4 argues is fatal to any level specification in this setting. A null from an unidentified specification says nothing in either direction. What the module establishes, and what is worth having, is that the within-firm green share is flat in the raw conditional means, which is at least consistent with the composition null.

5.8 How much the depth control matters

Environmental depth and general agreement depth correlate at 0.96 within destination and year. Table 9 varies the depth control across four alternatives: none, aggregate TotalDepth (17 areas), a version restricted to the 14 non-environmental areas most correlated with EP content, and the independently coded DESTA index (Dür, Baccini, and Elsig, 2014).

Table 9: EP×green under alternative depth controls (collapsed panel, WB index)

Depth control	EP×green	Asymptotic 95% CI	p
None	-0.0057	$[-0.0118,\ +0.0003]$	0.06
TotalDepth (17 areas) *[baseline]*	-0.0046	$[-0.0182,\ +0.0091]$	0.51
TotalDepth targeted (14 areas)	-0.0033	$[-0.0188,\ +0.0123]$	0.64
DESTA index	-0.0043	$[-0.0129,\ +0.0042]$	0.30
Product–dest FE		Yes	
Dest–year FE		Yes	
Product–year FE		Yes	
Cells		3,681,023	
Clusters		228 destinations	

Collapsed panel, weighted by cell size. Asymptotic intervals. The targeted control drops three WB areas with zero variance in Chinese agreements (labour-market regulation, visa and asylum, subsidies). DESTA covers seven policy areas coded independently of the World Bank instrument and excludes environmental provisions by construction, reducing the VIF from 5.7 to 1.9 and leaving the coefficient essentially unchanged.

The point estimate moves within a band of 0.0024 log points — narrower than a single standard error — stays negative throughout, and every interval contains zero. Dropping the depth control altogether moves the coefficient away from zero rather than toward it. The green null does not rest on the choice of depth control.

5.9 Environmental share of agreement content

The main specification estimates the marginal effect of one additional provision. Abman, Lundberg, and Ruta (2024) pose a different question: given that an agreement exists, does a more environmentally weighted agreement move composition? That framing requires restricting to partner destinations and replacing EP depth with its share of total depth, EP_{dt}/TD_{dt}. On the collapsed panel restricted to the 23 partner destinations (534,846 cells, 516,684 surviving singleton removal; the share takes 12 distinct values in $[0.012, 0.068]$; coefficient of variation 0.19 versus 0.62 for the level), the green interaction is -2.25 (s.e. 1.15, $p = 0.06$) and the dirty interaction -1.60 (s.e. 1.55, $p = 0.32$). The share varies far less than the level, so the standard errors are correspondingly wide; the green coefficient clears neither conventional threshold. It is reported because the contrast is the one used by the closest antecedent in the literature, and because a marginally negative coefficient is the opposite of what green trade creation would predict — not because this design identifies it well.

5.10 Sample robustness

Table 10 collects the full-panel variations. Adding controls (MFN tariff, market concentration, antidumping exposure) moves the green coefficient to an exact zero (-0.0002, $p = 0.94$). Excluding the eleven ASEAN destinations — the largest single cohort — leaves it at -0.0025 ($p = 0.48$). Re-including Hong Kong and Macao gives -0.0009 ($p = 0.80$). Across every variation, the dirty coefficient sits between -0.0040 and -0.0060 with asymptotic p-values of 0.02–0.09: persistently marginal under asymptotics, robust to none of the same battery applied to the green margin.

On the tariff control: the variable is the applied duty recorded in the customs data, not a rate constructed here; it varies across destinations for the same HS6–year in 97.9% of product–year cells and is therefore a destination-facing tariff, not a Chinese import duty. Its mean does not fall after PTA entry for partner destinations, which indicates it is the MFN rather than the preferential rate. The preferential rate was not obtainable. The headline estimates do not depend on the tariff (it appears only in the "with controls" row and in the CEM matching), and whatever liberalization the agreements delivered varies at the destination–year level and is absorbed by θ_{fdt}. The residual threat runs in the conservative direction: if

Table 10: Sample robustness (full firm-level panel, WB index)

		EP×green		EP×dirty	
Sample variation	Obs.	Coeff.	p	Coeff.	p
Baseline	21.5M	−0.0023	0.57	−0.0044	0.052
With controls	19.3M	−0.0002	0.94	−0.0060	0.022
Excluding ASEAN	19.2M	−0.0025	0.48	−0.0044	0.051
Including HK–Macao	23.6M	−0.0009	0.80	−0.0041	0.094
Common support	21.5M	−0.0022	0.57	−0.0044	0.052
PPML, zero-filled grid	7.9M	+0.0015	0.74	−0.0301	0.06
Within-firm green share	13.3M	−0.0001	0.50	—	
Firm–product–dest FE		Yes (all rows)			
Firm–dest–year FE		Yes (all rows)			
Product–year FE		Yes (all rows)			

WB index throughout. Asymptotic destination-clustered p-values; the dirty column's marginal significance does not survive wild cluster bootstrap, permutation, or leave-one-out inference (Section 5.4). Controls: $\ln(1 + \text{MFN tariff})$, $\ln \text{HHI}$, antidumping indicator. PPML estimated on a zero-filled HS6–dest–year grid (8.2M cells, excl. HK–Macao). Within-firm green share specification uses firm–destination and year FE (not triple-difference); see Section 5 for interpretation.

PTA tariff cuts fell more on green goods, the omitted preferential margin would load onto EP×green with a positive sign. The estimate is zero.

5.11 Destination-specific trends

The central falsification test asks whether allowing each destination its own linear trend in the green (and dirty) gap changes the conclusion, given that the conservative-bias argument of Section 4 implies those trends might be correlated with EP depth. Two variants bracket the answer, estimated on the collapsed panel.

The first adds destination-specific linear trends interacted with green and dirty status over the full sample period. WB results are essentially unchanged (EP×green −0.0051, bootstrap $p = 0.50$; EP×dirty −0.0082, bootstrap $p = 0.28$). The TREND green interaction flips sign and — the only instance in the full robustness battery — survives the wild cluster bootstrap ($p = 0.012$). Taken at face value it would say environmental provisions reduce green exports. It should not be taken at face value. Unit-specific trends estimated over the full sample period absorb post-treatment dynamics and can reverse the sign of treatment coefficients when effects or cohort dynamics are present (Wolfers, 2006); the late negative drift that these trends pick up is precisely the early-cohort artifact that the Sun–Abraham estimator already identifies as noise.

The second variant estimates destination-specific slopes on *pre-treatment* years only and projects them forward before re-estimating the main specification on the detrended outcome. Every coefficient returns to an imprecise zero: TREND×green +0.0074, bootstrap $p = 0.18$; WB×green +0.0168, bootstrap $p = 0.71$; dirty interactions $p = 0.30$–0.62. That the one nominally robust coefficient in the full-period trend exercise reverses sign entirely depending on how the trends are estimated identifies it as an artifact of the trend specification. The composition null stands against destination-specific linear drifts in green demand, the most natural form of the selection-on-trends confounder.

5.12 CO_2 intensity as a continuous dirty measure

The binary dirty indicator groups HS6 products into six classic pollution-intensive sectors. A continuous intensity measure might tell a different story. I construct one from the replication data of Shapiro (2021), which reports China-specific CO_2 emission intensities for 47 EXIOBASE industries. HS6 products are mapped to ISIC3 divisions via the same HS1996–ISIC3 concordance used for the binary classification; unmatched codes (9.5% of the panel) are assigned the sample mean. The resulting measure validates against the existing trichotomy without circularity: mean intensity is 0.0012 for dirty products, 0.0008 for neutral, and 0.0006 for green, monotonic in the expected direction.

Re-estimating equation (1) on the collapsed panel with the standardized intensity in place of the binary dirty indicator gives EP×intensity $= -0.0025$ (WB, bootstrap $p = 0.71$) and -0.0013 (TREND, asymptotic $p = 0.012$, bootstrap $p = 0.06$). The continuous measure fades under robust inference the same way the binary one does.

5.13 Alternative outcomes: quantity, unit value, and trimming

Log export value conflates shipment volume and price per unit. If EPs redirect composition without moving average prices — or vice versa — the value estimate would be an artifact of two channels canceling. Table 11 re-estimates equation (1) on the collapsed panel for three outcome variables: log export value (the main specification), log export quantity, and log unit value.

No outcome registers a significant effect on either index or either product type. The null is consistent across value, quantity, and price — it is not an artifact of channels canceling in the value aggregate.

Trimming. Trimming log export value at the 1st and 99th percentiles removes 2.0% of collapsed-panel cells (3,773,498 → 3,698,033); after singleton removal the estimation sample

Table 11: EP×green and EP×dirty across outcome variables (collapsed panel)

		WB index		TREND index	
Outcome	Interaction	Coeff.	Boot. p	Coeff.	Boot. p
Log export value	×green	−0.0046	0.65	−0.0002	0.87
	×dirty	−0.0119	0.07	−0.0015	0.86
Log export quantity	×green	−0.0055	0.53	+0.0019	0.43
	×dirty	−0.0115	0.25	−0.0004	0.89
Log unit value	×green	$p \in [0.85, 0.95]$		$p \in [0.85, 0.95]$	
	×dirty	$p \in [0.85, 0.95]$		$p = 0.16$ (TREND)	

Collapsed panel; FE $pd + dt + pt$; destination-clustered wild cluster bootstrap (B = 9,999). Log export value row is the baseline collapsed-panel estimate from Table 5. Unit-value coefficients are small in magnitude and reported as p-value ranges; exact coefficients available from the author. Asymptotic p-values for log unit value are between 0.85 and 0.95 on all four interactions except TREND×dirty ($p = 0.16$).

is 3,605,798. The green null is unaffected: EP×green −0.0048 (WB, bootstrap $p = 0.61$) and +0.0018 (TREND, bootstrap $p = 0.41$). The dirty margin reproduces its characteristic pattern — marginally significant under WB ($p = 0.041$), null under TREND ($p = 0.89$) — consistent with a false positive rather than a genuine effect.

6 Conclusion

China's trade agreements did not shift the composition of its exports between 2000 and 2015. On the green margin, the finding is a *bounded null*: stable across six estimation designs and nine sample variations, robust to bootstrap and permutation inference, and bounded from above by roughly one quarter of the reference estimate in the aggregate literature. This is not a failure to reject. It is a precisely circumscribed statement about the absence of a specific effect — the kind the literature has documented elsewhere under different conditions.

On the dirty margin, the finding is a *non-result*. A coefficient that is overwhelming under asymptotic standard errors survives none of the inference battery applied to the green margin. The leave-one-out exercise reveals why: the point estimate is stable across all 23 exclusions (−0.0097 to −0.0133), but the standard error nearly triples when either of two destinations is dropped. Precision, not the estimate, is what two specific destinations supply. An identified effect does not behave this way.

The central reading is about content. Environmental provisions affect trade when they are specific, binding, and targeted at a product category — Abman, Lundberg, and Ruta (2024) show this for deforestation, Brandi et al. (2020) for trade composition. China's chapters of this period were thin and cooperation-heavy: the two WB sub-indices with a direct trade mechanism are perfectly collinear in-sample and appear in only three treated country–years. The policy implication is direct: a chapter in an agreement is not itself the instrument. What the chapter contains is.

Methodologically, the paper adds a cautionary case. With treatment concentrated across roughly a dozen agreements, asymptotically significant composition effects can emerge and dissolve under honest inference. Bootstrap, permutation, and leave-one-out — applied not as robustness appendices but as the primary mode of inference — should be standard practice for this class of design.

Three limitations bear noting. The preferential tariff rate was not obtainable; the headline estimates are independent of it, but the MFN rate used as a control is an imperfect proxy. A continuous-dose estimator in the spirit of Callaway, Goodman-Bacon, and Sant'Anna (2024) is not implemented; the TWFE coefficient is a weighted average of cohort-specific effects, though the permutation test is agnostic about this weighting. The extensive margin is probed at the product–destination level: new firm entry into a destination where other Chinese firms already export the same green product is not captured.

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A Equivalence of the collapsed and firm-level panels

The collapsed panel aggregates firm–HS6–destination–year observations to HS6–destination–year cells with outcome equal to the within-cell mean of log exports and weights equal to cell counts. Under fixed effects $pd + dt + pt$ the weighted cell-level regression is algebraically identical to the unweighted micro-level regression under the same fixed effects. Verification to seven significant figures: the EP×green coefficient is -0.0045685 in both; the EP×dirty coefficient is -0.011873 in both. The result is exact up to floating-point precision and holds by construction of the collapse and weights, not by approximation.

B Saturation ladder

Table 12 reports EP-depth coefficients on log exports across twelve fixed-effects structures, from the sparsest (product–destination and year) to the most saturated (adding firm–destination–year and product–year). Moving from left to right, the coefficient falls monotonically, reaching a precise zero in the three most saturated structures. This is the empirical documentation of the collinearity argument of Section 4: the pattern is the signature of selection, not treatment.

Table 12: Why the level effect is not identified: saturation ladder

	WB EP depth		*TREND EP count*	
Fixed effects	(1) Baseline	(2) Controls	(3) Baseline	(4) Controls
$fpd + t$	0.00311	0.00354	0.00040	0.00046
	(0.00251)	(0.00260)	(0.00047)	(0.00048)
$fpt + pd$	0.00463*	0.00476*	0.00101**	0.00105**
	(0.00270)	(0.00274)	(0.00051)	(0.00052)
$fpt + fpd$	0.00010	0.00010	0.00016	0.00016
	(0.00326)	(0.00329)	(0.00061)	(0.00060)
$fpd + pt$	0.00087	0.00103	0.00028	0.00030
	(0.00225)	(0.00221)	(0.00044)	(0.00044)

Dependent variable: log export value. Each row adds richer fixed effects, progressively eliminating less credible variation. The EP depth coefficient is small throughout and nominally significant only at intermediate structures, disappearing once firm–dest–year absorption is included — the signature of selection into agreements rather than a treatment effect. f: firm, p: product, d: destination, t: year. Standard errors clustered by destination. Controls in columns (2) and (4): non-environmental TotalDepth. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Sun–Abraham event study

Figure 2 and Table 13 report the interaction-weighted (L. Sun and Abraham, 2021) event study of the green and dirty versus neutral differential, estimated on the destination-year composition gap and run with `eventstudyinteract` in Stata.

A note on standard errors. The cohort shares with which `eventstudyinteract` aggregates group-specific effects are themselves estimated from the data; Sun and Abraham (2021) prescribe including that estimation uncertainty in the variance. `eventstudyinteract` does so; `fixest::sunab` in R treats the shares as fixed weights. The two implementations agree on coefficients to 15 significant figures but disagree on standard errors. The degree of disagreement is informative: where a single cohort identifies a period (so there are no shares to estimate), the ratio of the two standard errors is exactly 1.00; where many cohorts contribute and their effects are heterogeneous, the ratio reaches 3 to 4. The standard errors reported here are the Stata ones, which reflect the full estimation uncertainty.

With these standard errors, no coefficient in the $[-10, +8]$ window is individually distinguishable from zero on the green or dirty margin. The ATT is -0.042 ($p = 0.27$) for the green gap and $+0.073$ ($p = 0.28$) for the dirty gap. Any apparent pre-trend in the R output (the lead at $t = -6$ had $p = 0.001$ under `fixest::sunab`) reflects the treatment of cohort-share uncertainty, not a genuine pre-trend anomaly.

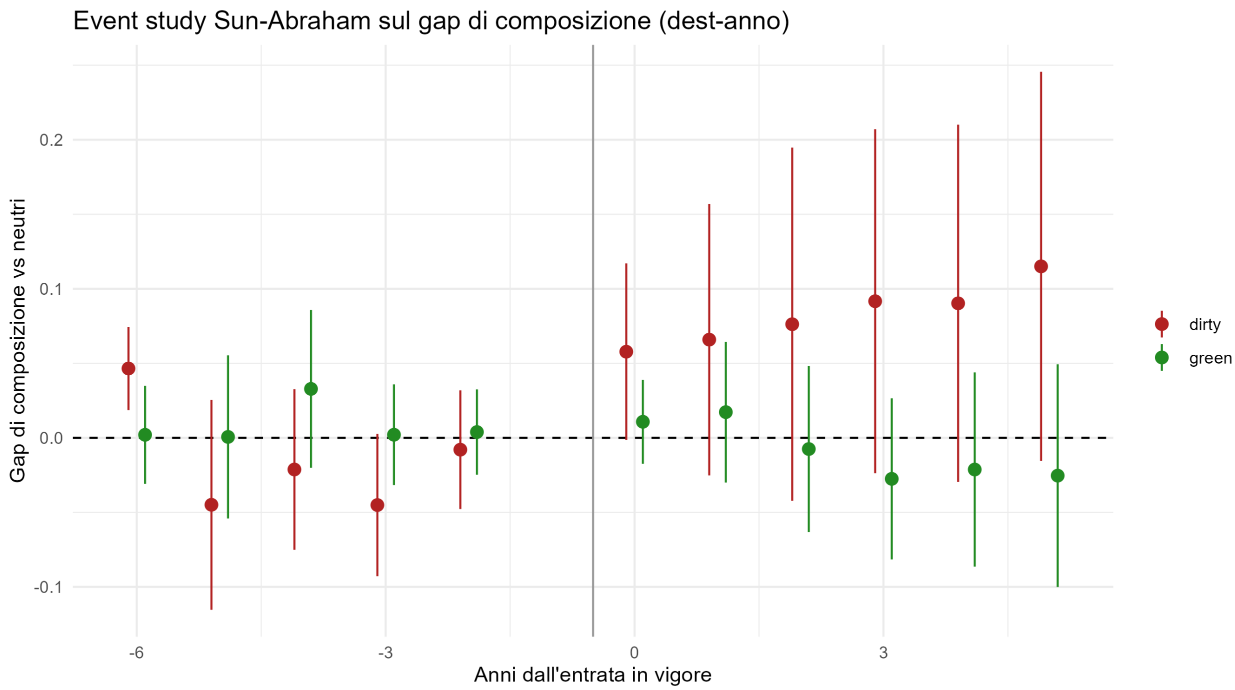

Figure 2: Sun–Abraham interaction-weighted event study: green and dirty vs. neutral gap at the destination–year level. 95% confidence bands based on `eventstudyinteract` standard errors, which account for estimation uncertainty in cohort shares. Endpoint bins at $t \leq -10$ and $t \geq +8$.

D Sub-index estimates

Table 14 reports triple-difference estimates replacing the aggregate EP index with each sub-index in turn. Each row is a separate regression; the coefficients are not partial effects. See Section 5.5 for interpretation.

Table 13: Sun–Abraham interaction-weighted event study: composition gap estimates

Years since entry	Green − neutral gap Coeff.	p	Dirty − neutral gap Coeff.	p
−10	−0.0500	0.419	−0.0356	0.598
−9	−0.0128	0.881	0.0641	0.424
−8	−0.0001	0.999	−0.0478	0.436
−7	−0.0392	0.422	−0.0186	0.561
−6	0.0027	0.959	0.0466	0.335
−5	0.0022	0.961	−0.0447	0.378
−4	0.0345	0.359	−0.0210	0.538
−3	0.0021	0.938	−0.0451	0.189
−2	0.0034	0.861	−0.0080	0.832
+0	0.0116	0.504	0.0579	0.107
+1	0.0191	0.470	0.0661	0.192
+2	−0.0049	0.881	0.0765	0.241
+3	−0.0244	0.501	0.0921	0.156
+4	−0.0187	0.711	0.0906	0.181
+5	−0.0233	0.661	0.1153	0.117
+6	0.0027	0.959	0.1264	0.165
+7	−0.0320	0.541	0.1083	0.223
+8	−0.0779	0.181	0.0983	0.235
Aggregated ATT	−0.0421	0.269	0.0729	0.276

Estimated via `eventstudyinteract` in STATA (). Dependent variable: destination–year composition gap (mean log exports of green or dirty products minus mean log exports of neutral products, within destination–year cells). Standard errors propagate estimation uncertainty in cohort shares, as prescribed by L. Sun and Abraham (2021); `fixest::sunab` in R treats the shares as known weights and yields narrower standard errors — the ratio reaches 3–4 in periods identified by many discordant cohorts. Coefficients agree to 15 significant figures across implementations. Endpoint bins at $t \leq -10$ and $t \geq +8$ accumulate all observations beyond those thresholds. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 14: Provision sub-indices: which clauses have a trade mechanism?

Sub-index	×green	×dirty
Mechanism-bearing sub-indices		
Green goods liberalization (WB)	-0.0115	-0.0876^{*}
	(0.0879)	(0.0483)
	[0.896]	[0.071]
Green market access (TREND)	-0.0271	-0.0491^{**}
	(0.0303)	(0.0243)
	[0.372]	[0.045]
Enforcement		
Dispute settlement mechanism (WB)	-0.0048	0.0067
	(0.0394)	(0.0525)
	[0.903]	[0.898]
Dispute settlement mechanism (TREND)	0.0042	-0.0052
	(0.0148)	(0.0140)
	[0.778]	[0.709]
Binding obligations (TREND)	0.0022	-0.0013
	(0.0047)	(0.0037)
	[0.638]	[0.724]
Placebo sub-indices (no trade mechanism)		
Cooperation-only provisions (TREND)	0.0132	0.0019
	(0.0121)	(0.0108)
	[0.275]	[0.858]
Regulatory space clauses (TREND)	0.0242^{**}	0.0225^{***}
	(0.0099)	(0.0086)
	[0.015]	[0.010]
Cells (all rows)	3,681,023	

Baseline variant, collapsed panel. Each row is a separate regression replacing the aggregate EP index with the indicated sub-index; TotalDepth controls remain included. Standard errors in parentheses; asymptotic p-values in brackets. The two mechanism-bearing WB sub-indices are perfectly collinear in-sample and non-zero in only three treated country–years (Korea from 2015, Switzerland from 2014), so their coefficients are not identified separately from each other. The regulatory space clauses, despite surviving wild cluster bootstrap, cannot be distinguished from general agreement depth (within-sample correlation with TotalDepth: 0.90); see Section 5.5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.